

IoT in Robotics & Automation

An educational overview of connected intelligence, autonomous systems, and the future of smart machines.

What is IoT in Robotics?

The Internet of Things (IoT) brings robotics into a connected world — where machines don't just act, they **sense, communicate, and respond** intelligently.

Real-Time Data

Connected devices collect and share data instantly across networks

Robot-to-Cloud

Robots communicate with each other and cloud systems seamlessly

Remote Intelligence

Enables remote monitoring and intelligent decision-making at scale

Role of AI in Autonomous Robots

Perception

Computer vision and sensors interpret the robot's surroundings in real time

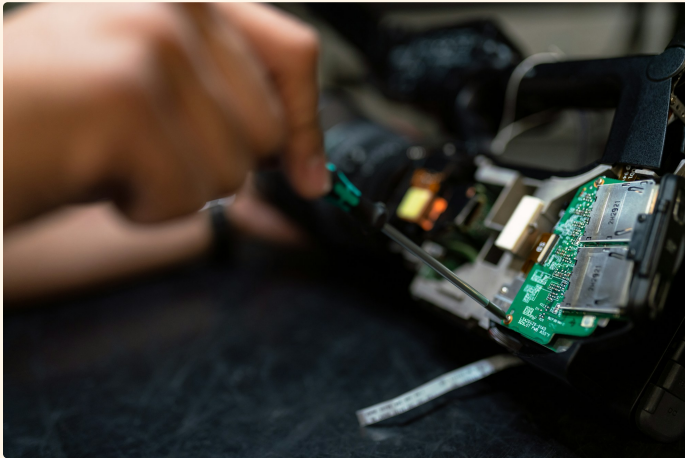
Decision-Making

Machine learning algorithms evaluate options and choose the optimal action

Adaptation

AI learns from experience over time to continuously improve performance

PLC-Based Automation System Design



What is a PLC?

Programmable Logic Controllers (PLCs) are rugged industrial computers that serve as the brain of automated machinery — executing logic, managing inputs, and driving outputs with precision.

1 Industrial Control

PLCs manage machinery across factories and production lines

2 Input / Output

Sensors feed data in; actuators execute commands out

3 Real-Time Precision

Precise timing for manufacturing and assembly operations

Key Components of PLC Systems



Sensors

Temperature, pressure, and proximity sensors gather live environmental data



Actuators

Motors, valves, and solenoids convert signals into physical action



Communication

Ethernet and Modbus networks link PLCs to supervisory systems



Humanoid Robot Design Concepts

Humanoid robots are engineered to replicate human form and function — combining mechanics, electronics, and AI into a single, adaptable machine.

1

Bipedal Locomotion

Walking on two legs with dynamic balance and stability control systems

2

Dexterous Hands

Multi-jointed fingers enabling precise grasping and manipulation tasks

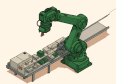
3

Sensor Integration

Vision, touch, and proprioception work together for full situational awareness



Real-World Applications



Manufacturing

Automated assembly lines with integrated quality control and zero-downtime production



Healthcare

Surgical robots delivering precision procedures and robotic assistants supporting patient care



Logistics

Autonomous warehouse robots and delivery systems revolutionizing supply chains